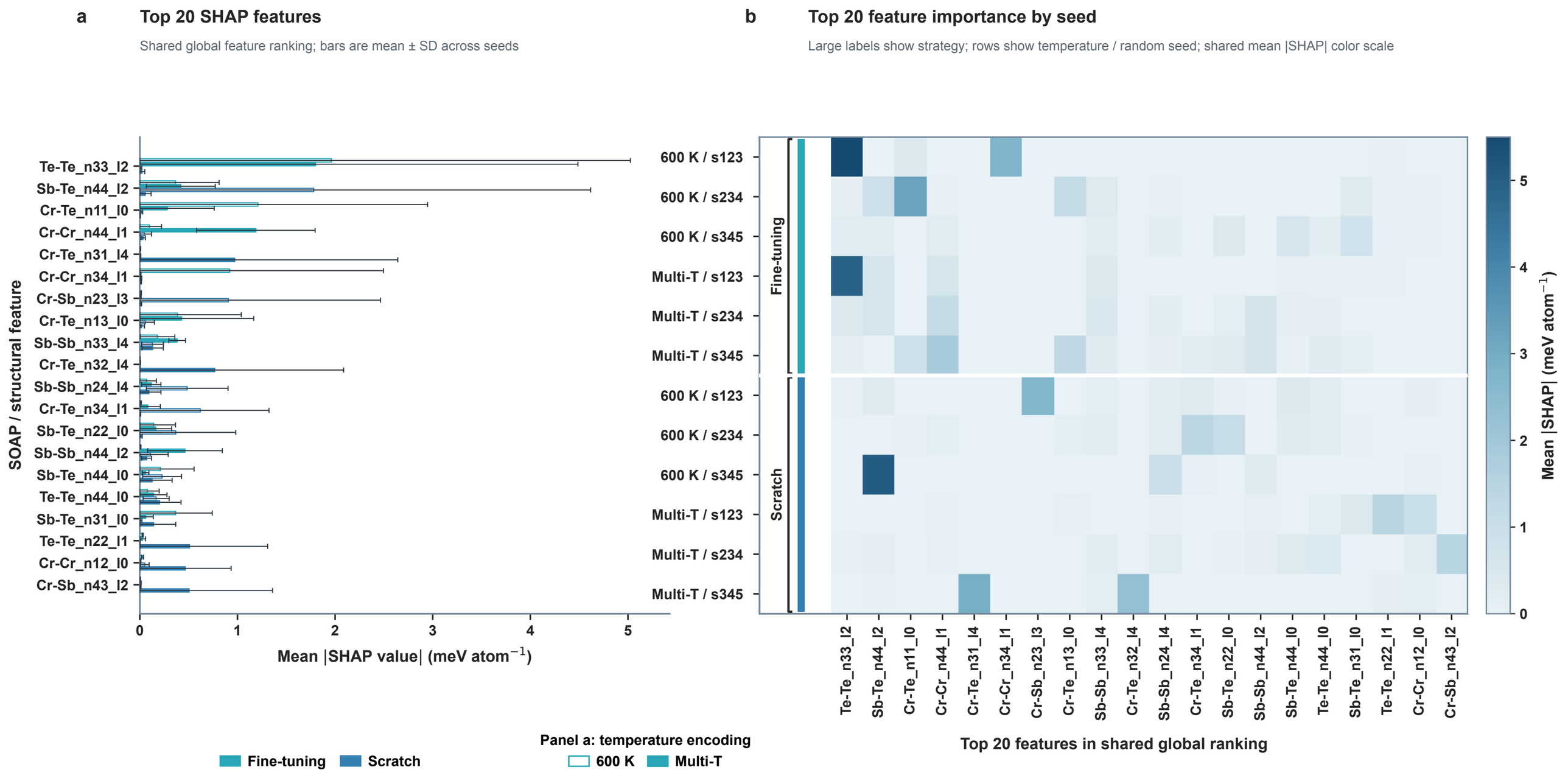

a
Top 20 SHAP features
Shared global feature ranking; bars are mean ± SD across seeds
Te-Te_n33_l2
Sb-Te_n44_l2
Cr-Te_n11_l0
Cr-Cr_n44_l1
Cr-Te_n31_l4
Cr-Cr_n34_l1
Cr-Sb_n23_l3
Cr-Te_n13_l0
Sb-Sb_n33_l4
Cr-Te_n32_l4
Sb-Sb_n24_l4
Cr-Te_n34_l1
Sb-Te_n22_l0
Sb-Sb_n44_l2
Sb-Te_n44_l0
Te-Te_n44_l0
Sb-Te_n31_l0
Te-Te_n22_l1
Cr-Cr_n12_l0
Cr-Sb_n43_l2
SOAP / structural feature
0
1
2
3
4
5
Mean |SHAP value| (meV atom^{-1})
b
Top 20 feature importance by seed
Large labels show strategy; rows show temperature / random seed; shared mean |SHAP| color scale
Fine-tuning
Scratch
600 K / s123
600 K / s234
600 K / s345
Multi-T / s123
Multi-T / s234
Multi-T / s345
Mean |SHAP| (meV atom^{-1})
Top 20 features in shared global ranking
Panel a: temperature encoding
600 K
Multi-T
Fine-tuning
Scratch